

Structure for IIR systems

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Structure for IIR systems:

LTI discrete time systems can be characterized by Linear constant coefficient difference equation(LCCDE)

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

Zeros

Poles

Direct Form Structures:

Direct Form I

Direct Form II

Transposed

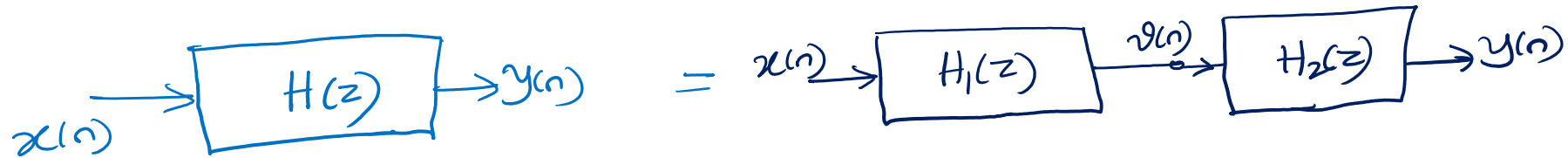
Cascaded

Parallel

Lattice-Ladder structure

Direct Form I structure realization:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \sum_{k=0}^M b_k z^{-k} \times \frac{1}{1 + \sum_{k=1}^N a_k z^{-k}} = H_1(z) H_2(z) \quad \text{--- ①}$$



$$\text{i.e., } H(z) = \frac{Y(z)}{X(z)} = \frac{Y(z)}{X(z)} \cdot \frac{Y(z)}{Y(z)} \quad \text{--- ②}$$
$$= H_1(z) \cdot H_2(z)$$

$$H_1(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^M b_k z^{-k} \quad \text{--- ③} \quad \text{and} \quad H_2(z) = \frac{1}{1 + \sum_{k=1}^N a_k z^{-k}} \quad \text{--- ④}$$

Consider eqn (3)

$$H_1(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^M b_k \bar{z}^k \quad \text{--- (3)}$$

$$Y(z) = \sum_{k=0}^M b_k X(z) \bar{z}^k$$

Taking inverse Z.T. we get

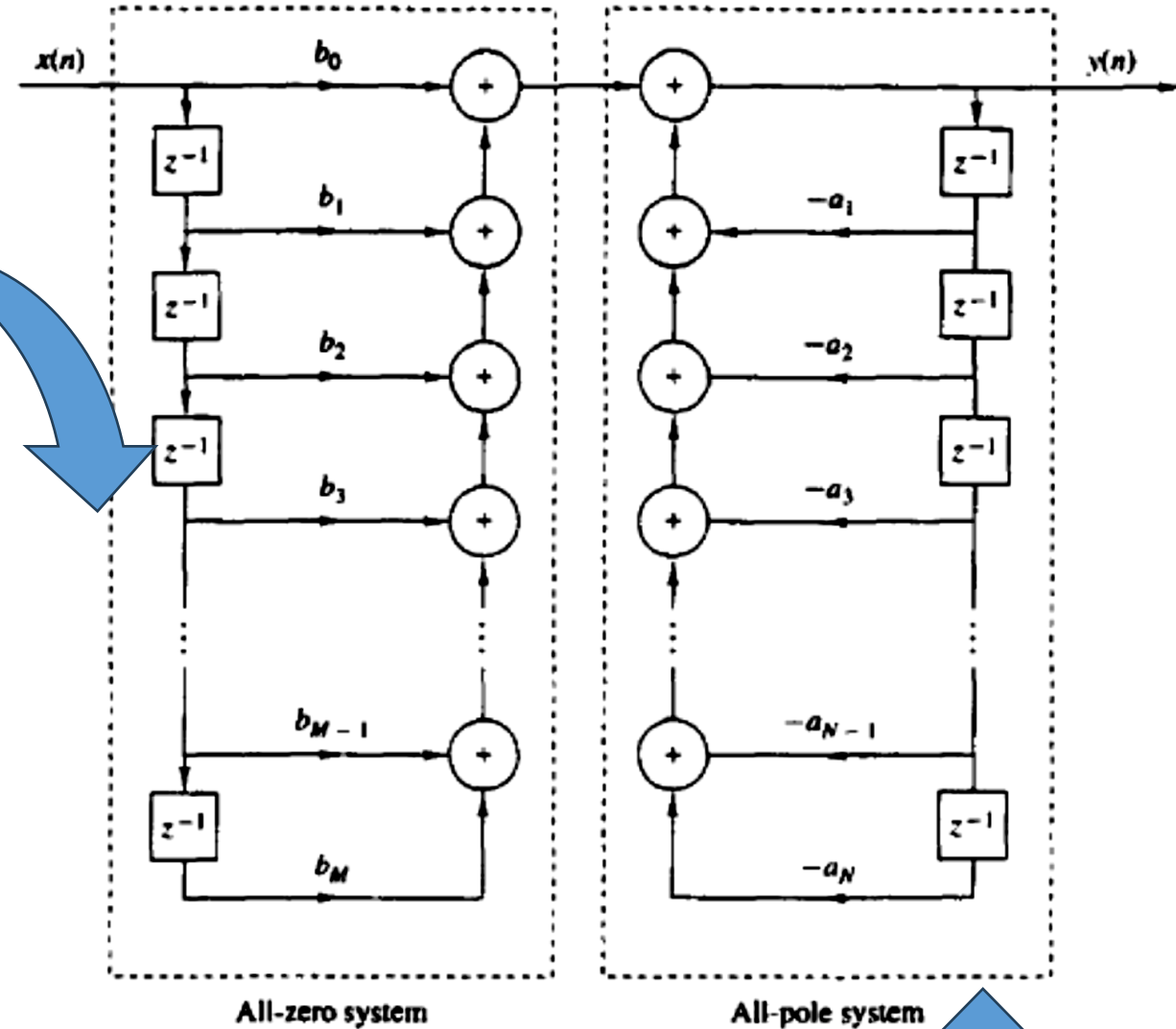
$$y(n) = \sum_{k=0}^M b_k x(n-k) \quad \text{--- (5)}$$

Consider eqn (4) $H_2(z) = \frac{Y(z)}{V(z)} = \frac{1}{1 + \sum_{k=1}^N a_k \bar{z}^k}$

$$Y(z) + \sum_{k=1}^N a_k \bar{z}^k Y(z) = V(z)$$

$$Y(z) = V(z) - \sum_{k=1}^N a_k \bar{z}^k Y(z)$$

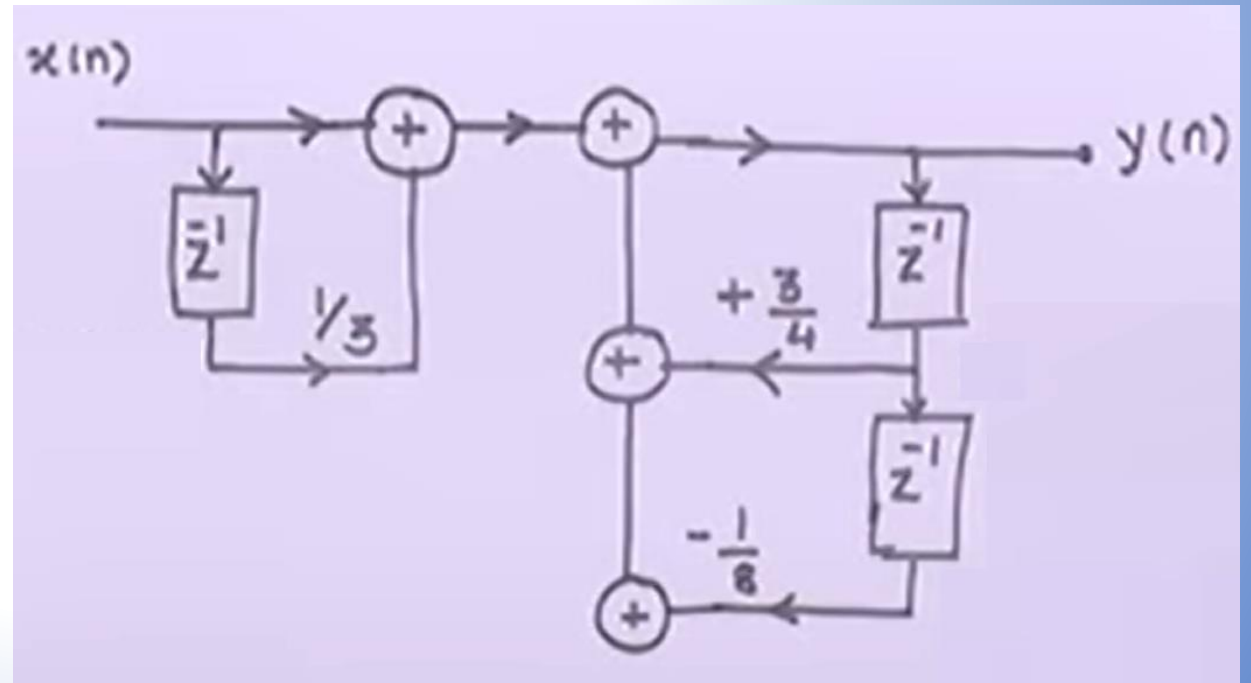
or $y(n) = v(n) - \sum_{k=1}^N a_k y(n-k) \quad \text{--- (6)}$



Find Direct form I structure

$$y(n) - \frac{3}{4} y(n-1] + \frac{1}{8} y(n-2) = x(n) + \frac{1}{3} x(n-1)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3} \bar{z}^{-1}}{1 - \frac{3}{4} \bar{z}^{-1} + \frac{1}{8} \bar{z}^{-2}}$$



Direct Form II structure realization:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{1}{1 + \sum_{k=1}^N a_k z^{-k}} \times \sum_{k=0}^M b_k z^{-k} = H_1(z) H_2(z)$$

$$H_1(z) = \frac{V(z)}{X(z)} = \frac{1}{1 + \sum_{k=1}^N a_k z^{-k}}$$

and

$$H_2(z) = \frac{Y(z)}{V(z)} = \sum_{k=0}^M b_k z^{-k}$$

$$V(z) + \sum_{k=1}^N a_k z^{-k} V(z) = X(z)$$

$$Y(z) = \sum_{k=0}^M b_k z^{-k} V(z)$$

Taking inverse Z.T.

$$v(n) = x(n) - \sum_{k=1}^N a_k v(n-k)$$

$$y(n) = \sum_{k=0}^M b_k v(n-k)$$

Let us rewrite

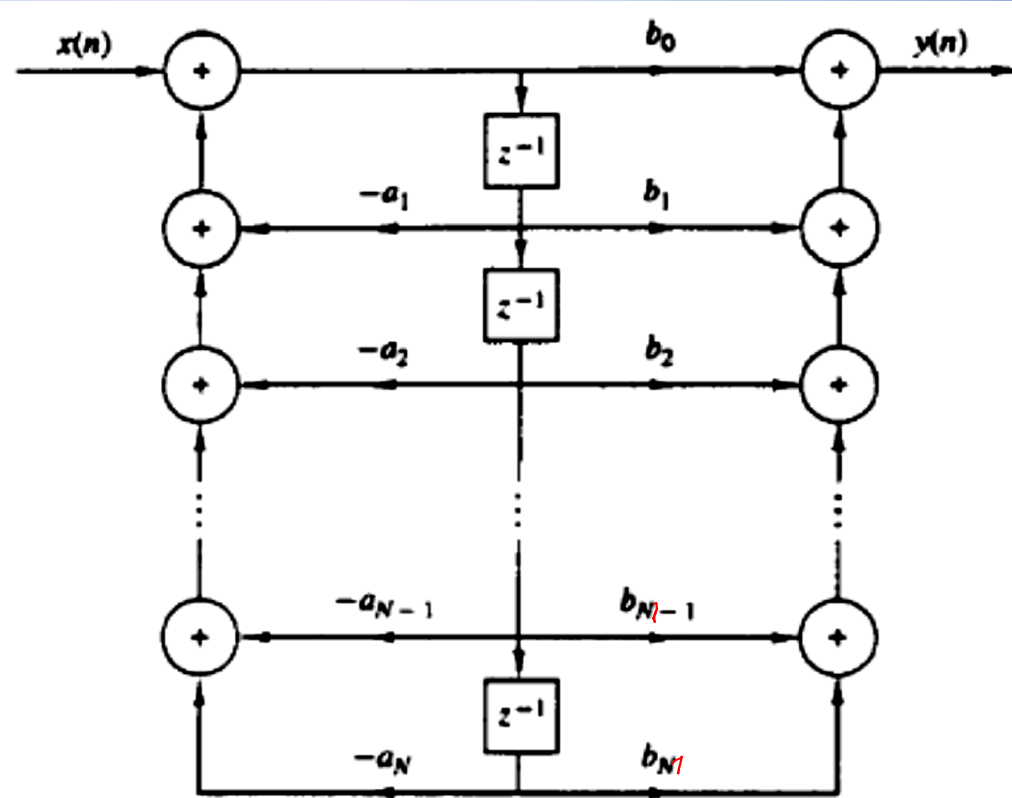
$$v(n) = x(n) - \sum_{k=1}^N a_k v(n-k)$$

$$v(n) = x(n) - a_1 v(n-1) - a_2 v(n-2) \dots - a_N v(n-N)$$

$$y(n) = \sum_{k=0}^M b_k v(n-k)$$

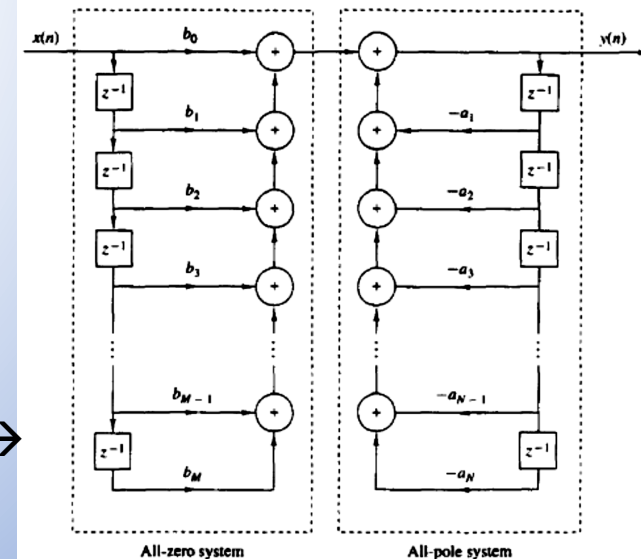
$$y(n) = b_0 v(n) + b_1 v(n-1) + \dots + b_M v(n-M)$$

DF 2 →



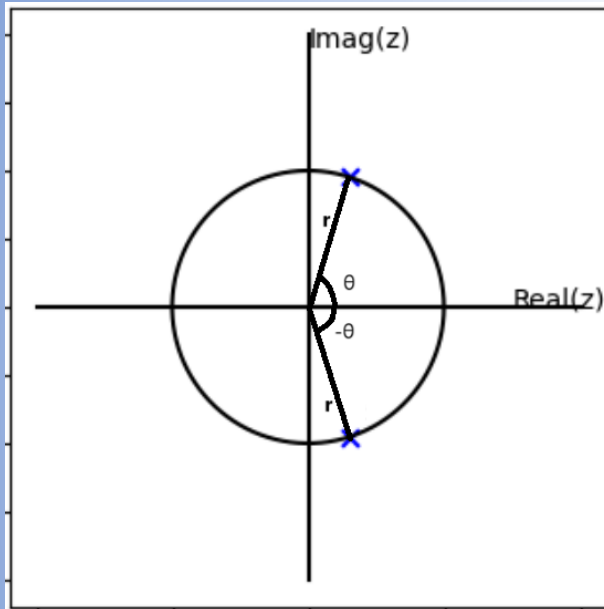
Delay blocks are reduced –
Canonic form

DF 1 →



Two pole structures:

- If zero/pole is complex – difficult to implement (complex coefficients)
- This can be converted into structure with real coefficients – 2 pole structures



$$H(z) = \frac{1}{(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1})}$$

$$= \frac{1}{1 - (re^{j\theta} + re^{-j\theta})z^{-1} + r^2z^{-2}}$$

$$H(z) = \frac{1}{1 - 2r\cos\theta z^{-1} + r^2z^{-2}}$$

$$H(z) = \frac{1}{(1 - p_1z^{-1})(1 - p_1^*z^{-1})}$$

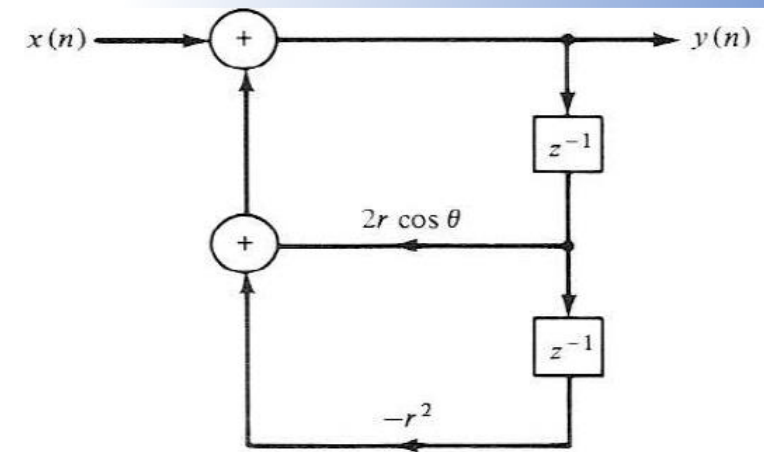


FIGURE 7.44 Realization of a two-pole IIR filter.

Recap – Goertzel algorithm:

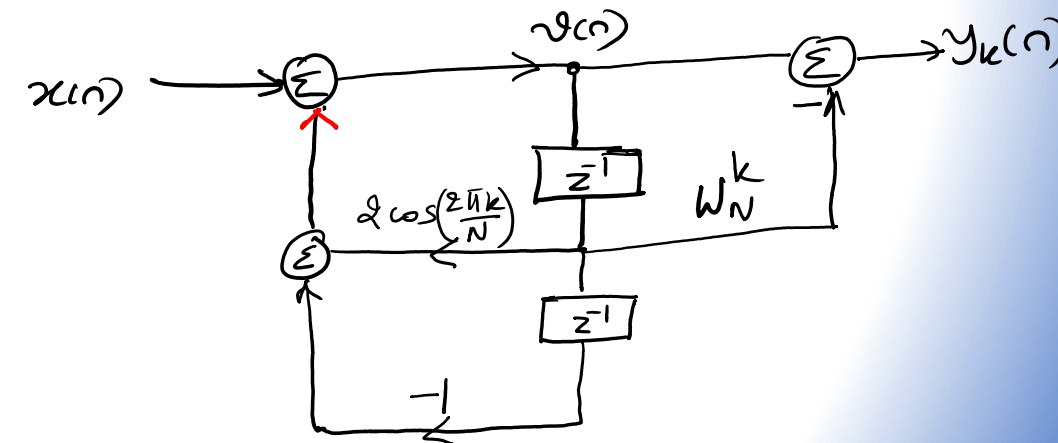
$$h(n) = W_N^{-nk} u(n) \Leftrightarrow H(z) = \frac{1}{1 - W_N^{-k} z^{-1}} \quad \text{--- (5)}$$

Now we can realize the structure for equation (5) as follows:

$$H_k(z) = \frac{1}{1 - W_N^{-k} z^{-1}} \frac{(1 - W_N^k z^{-1})}{(1 - W_N^k z^{-1})} = \frac{1 - W_N^k z^{-1}}{1 - (W_N^k + W_N^{-k}) z^{-1} + z^{-2}}$$

$$H_k(z) = \frac{1 - W_N^k z^{-1}}{1 - 2 \cos\left(\frac{2\pi k}{N}\right) z^{-1} + z^{-2}} \quad \text{--- (6)}$$

$$\text{Let } H_k(z) = \frac{Y_k(z)}{X(z)} = \frac{1 - W_N^k z^{-1}}{1 - 2 \cos\left(\frac{2\pi k}{N}\right) z^{-1} + z^{-2}}$$



Direct form II realization for computing k^{th} DFT point

*Thank
you*

